Labor Market

Intermediate Macroeconomics

Fall 2025, Sciences Po Campus du Havre

Lecture 5

Announcements

- Textbook: Chapter 7
- Quiz 1: The answer is now marked, and good job everyone!

Basics

Basic definitions

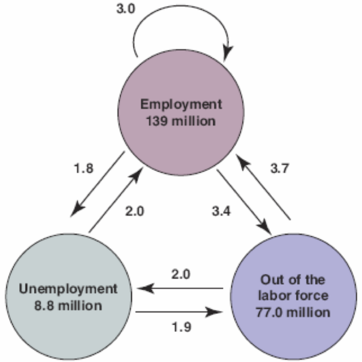
- Employment
- Unemployment
- Labor force
- Unemployment rate
- Labor Force Participation Rate
- Employment Rate

Basic definitions

- Employment (E): the number of people who have a job
- Unemployment (U): the number of people who do not have a job but are looking for one
- Labor force (L):
$$L = E + U$$
- Unemployment rate (u):
$$u = \frac{U}{L}$$
- Labor Force Participation Rate:
$$\frac{L}{Population(age16+)}$$
- Employment Rate:
$$\frac{E}{Population(age16+)}$$

Labor dynamics

Flows between Employment, Unemployment, and Non-participation in the US



1. The flows of workers in and out of employment are large.
2. The flows in and out of unemployment are large relative to the number of unemployed.
3. There are also large flows in and out of the labor force, much of it directly to and from employment.

Wage Determination

Wage Determination

Two perspectives

1. Bargaining
2. Efficiency

Wage Bargaining

- Employee and employer bargain to set the wage
- **Collective bargaining**: employee is represented by the labor union
- Sector-specific & country-specific
  - Slightly more than 10% of U.S. workers' wages are set by collective bargaining.
  - Collective bargaining plays an important role in Japan and most European countries.
- The higher the skills needed to do the job, the more likely there is to be bargaining between employers and individual employees.

Wage Bargaining

- **Reservation wage**: worker is indifferent between working or being unemployed
- if offered wage $\geq$ reservation wage $\implies$ worker is hired
- Wages typically depends on labor-market conditions (business-cycle):
  The lower the unemployment rate, the higher the wages.
- Bargaining power depends on:
  - How costly for the firm to find other workers
  - How hard for workers to find another job

Efficiency wage theories

Main idea: labor productivity is related to the wage

- Firms may want to pay a wage above the reservation wage in order to decrease workers' turnover and increase productivity.
- When unemployment is low, firms that want to avoid an increase in quits will increase wages to induce workers to stay with the firms.

Model

Wage Setting (WS)

$$W = P^e F(u, z)$$

The aggregate nominal wage W depends on:

- Unemployment rate (u)
- Institutional and structural factors (z)
- Expected price level (P^e)

The aggregate nominal wage W depends on

- **Unemployment rate (u)**
  - When u is high, it is easy to replace the worker with others $\implies$ bargaining power is low
  - When u is high, workers are highly motivated to work and be productive $\implies$ efficiency wage motive
- **Institutional and structural factors (z)**
  - Level of minimum wage, employment protection, generosity of unemployment insurance etc.
  - Institutional and structural factors (z)
- **Expected price level (P^e)**
  - Both w and f care about **real wages**: $(\frac{W}{P})$
  - Why expected? Because wages do not adjust regularly enough. What matters are the expectations about the future price level
  - Simplification: $P^e = P$

Price Setting (PS)

1. Production function

- Recall our production function: $Y = F(K, AN)$
- Now we are in the medium run: K is constant and $K = 1$,
  A is constant and $A = 1$
- We can simplify the production function:

$$Y = N$$

- The cost of producing 1 more unit of output is equal to the cost of employing one more worker at wage W $\implies$ The marginal cost of production is equal to W

Price Setting (PS)

2. Mark-ups

- Now, let's assume that the markets are not perfectly competitive and firms have a market power:

$$P = (1 + m)W$$

- where m is a mark-up

Wage setting = Price setting

Wage setting

$$\frac{W}{P} = F(u, z)$$

- The higher the unemployment rate, the lower the real wage chosen by wage setters.
- The wage-setting relation is the relation between the real wage and the rate of unemployment

Price setting

$$\frac{W}{P} = \frac{1}{1+m}$$

Natural rate of unemployment

$$F(u, z) = \frac{1}{1+m}$$

The natural rate of unemployment (u_n): unemployment rate such that the real wage chosen in WS is equal to the real wage implied by PS.

At home (Based on Chapter 7-5)

1. Increase in unemployment benefits ($\nearrow z$)
2. Higher monopoly power: new antitrust law ($\nearrow m$)

Increase in unemployment benefits ($\nearrow z$)

WS shifts upwards $\implies$ higher unemployment rate

1. Higher unemployment benefits make the prospect of unemployment less painful
2. Wage-setters will require higher wage *given the existing unemployment rate*
3. WS shifts upwards
4. In equilibrium (WS=PS), the economy moves to the new state where the natural rate of unemployment increases ($u_n' > u_n$)

Higher monopoly power: new antitrust law ($\nearrow m$)

$$\nearrow m \implies \searrow \frac{1}{1+m} \implies \text{PS shifts downwards} \implies \text{higher unemployment rate}$$

1. $\nearrow m \implies$ All firms increase their prices
2. Nominal wages (W) do not change but due to higher prices real wages (W/P) decrease
3. PS shifts downwards
4. Higher unemployment is required for workers to accept the lower real wage
5. Movement along the WS curve until PS=WS
6. unemployment increases ($u_n' > u_n$)